

Topic 10 Summary: Centre of Gravity

- Definitions
 - Centre of Gravity
 - Centre of mass
 - Centroid
- Centre of gravity of a system of coplanar particles.
- Centre of gravity of simple 2-D bodies
- Centroids for curved areas
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Motivation

For every system and at every instant in time, there is a unique location in space that is the average position of the system's mass. This place is called the *center of mass*, commonly designated by cm, c.o.m., COM, G, c.g., or $\odot$.

One of the routine but important tasks of many real engineers is to find the center of mass of a complex machine^①. Just knowing the location of the center of mass of a car, for example, is enough to estimate whether it can be tipped over by maneuvers on level ground. The center of mass of a boat must be low enough for the boat to be stable. Any propulsive force on a space craft must be directed towards the center of mass in order to not induce rotations. Tracking the trajectory of the center of mass of an exploding plane can determine whether or not it was hit by a massive object. Any rotating piece of machinery must have its center of mass on the axis of rotation if it is not to cause much vibration.

Also, many calculations in mechanics are greatly simplified by making use of a system's center of mass. In particular, the whole complicated distribution of near-earth gravity forces on a body is equivalent to a single force at the body's center of mass. Many of the important quantities in dynamics are similarly simplified using the center of mass.

^① Nowadays this routine work is often done with CAD (computer aided design) software. But an engineer still needs to know the basic calculation skills, to make sanity checks on computer calculations if nothing else.

CENTRE OF GRAVITY

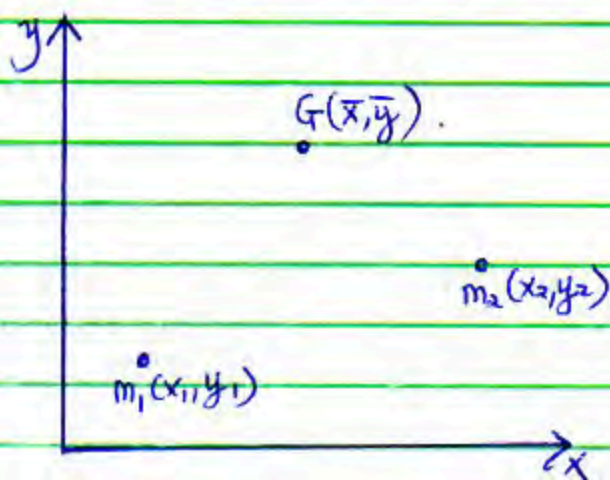
Any extended body is made up of many particles whose weight, unless the body is very large, forms a system of parallel forces. The resultant of these parallel forces passes through a fixed point for any orientation of the body. This point is the centre of gravity of the body.

If a body has a centre of symmetry G , such that for every particle A of the body, there is a corresponding particle A' of equal mass, and G is the mid-point of AA' , then the resultant weight of A and A' may be taken to act at G , and similarly for every other pair of particles, and hence G is the centre of gravity of the body.

Thus the CoG of a thin uniform rod is at its mid-point, that of a rectangle or parallelogram is at the intersection of the diagonals. For a circle, ellipse or sphere, at the centre of symmetry.

Centre of Gravity of a system of Coplanar Particles.

Consider a set of coplanar particles with masses $m_1, m_2, \dots$ and positions relative to a pair of x and y axes, of $(x_1, y_1), (x_2, y_2), \dots$. Let the total mass be M , and the Centre of gravity be the point $G(\bar{x}, \bar{y})$.



To obtain expressions for $\bar{x}$ and $\bar{y}$, we use the principle of moments. i.e.

moments of resultant = moments of original system.

Suppose the weights act at right angles to the plane of the paper, so that they have moments about both axes.

Applying the principle to the y-axis

$$(m_1 + m_2 + \dots) g \bar{x} = m_1 g x_1 + m_2 g x_2 + \dots$$

$$\therefore M \bar{x} = \sum m_i x_i$$

$$\Rightarrow \bar{x} = \frac{\sum m_i x_i}{M}$$

Taking moments about x-axis.

$$\bar{y} = \frac{\sum m_i y_i}{M}$$

Centre of Mass, Centroid

The centre of mass of a system of coplanar particles is defined by the above standard equations for $\bar{x}$ and $\bar{y}$. Since the g 's have cancelled out, these equations make no direct reference to weight, and they are meaningful whether or not the system or body is near a large planet and has weight, and whether or not the weights of the constituent particles are parallel. Centroid denotes a geometric centre and is defined to be coincident with the centre of gravity of a body provided that the matter in that body is evenly distributed. (is of equal density throughout)

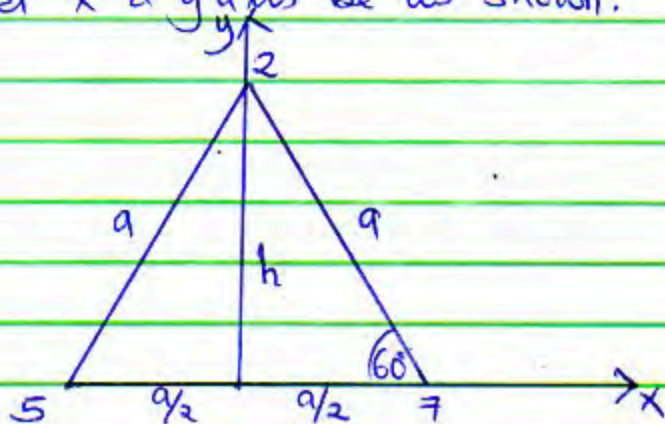
For uniform bodies of ordinary size near the surface of the earth, the 3 points are coincident.

Example 1:

Particles of mass 2kg, 5kg and 7kg are placed at the vertices of an equilateral triangle of side a .
Locate their centre of gravity.

Sol:

Let x & y axis be as shown.



Taking moments about y -axis

$$7\left(\frac{a}{2}\right) - 5\left(\frac{a}{2}\right) = 14\bar{x} \quad 14 = \text{total mass}$$

$$\therefore \bar{x} = \frac{a}{14}$$

$$h = a \sin 60 = \frac{a\sqrt{3}}{2}$$

Taking moments about x -axis

$$\frac{2a\sqrt{3}}{2} = 14\bar{y}$$

$$\bar{y} = \frac{a\sqrt{3}}{14}$$

$$\therefore \text{CoG, } G \left(\frac{a}{14}, \frac{a\sqrt{3}}{14} \right)$$

Exercise

1. Particles of mass $5m$, $10m$, $8m$ and $7m$ are placed at the points A, B, C , and D respectively of a straight line $ABCD$ in which $AB = BC = CD = a$. Find the distance of the centre of gravity from B . ($\frac{17}{30}a$)
2. Particles of mass $2m$, $3m$, $5m$, $10m$ are placed at the points $(1, 2)$, $(3, -1)$, $(-2, 0)$ and $(0, -3)$ respectively. Find $\bar{x}$ and $\bar{y}$. ($0.05, -1.45$)

Centre of Gravity of Simple 2-D Bodies.

Certain bodies have clear geometrical centres and if the matter in them is uniformly distributed, the centres of gravity of such bodies must lie at these points. Eg. uniform rod, uniform laminae like circles and rectangles.

The centre of gravity of a uniform triangular lamina lies at the intersection of its medians i.e. at the point which divides each median in the ratio $1:2$.

Three equal particles placed at the vertices of a triangle have the same centre of gravity as the triangular lamina.

To find the CoG of a compound body (e.g. rectangle + square, rectangle - triangle), we use the following principles.

a) In case of adding figures:

moment of whole = sum of moments of parts.

b) In case of remainders

moment of whole = moment of part removed + moment of remainder.

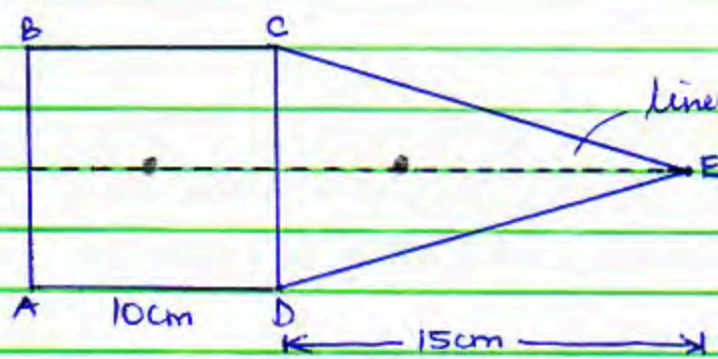
Note:-

If the laminae are represented as lying in the plane of the paper one is using, moments should be taken about axes which also lie in that plane.

Example 2:-

A uniform lamina ABCED consists of an isosceles triangle CED of height 15cm added to a square ABCD of side 10cm. Locate the centre of gravity.

Sol:-



line of symmetry

In other words
get $\bar{x}$ & find
cm between

Figure	Area	Distance of CoG from AB.
ABCD	100	5
CED	75	$10 + 15/3 = 15$
ABCED	175	X

Using:

Moment of whole = Sum of moments of parts

Taking moments about AB

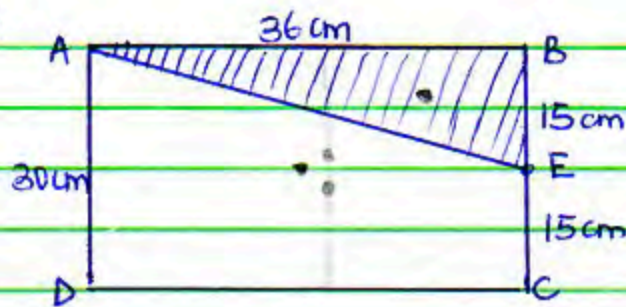
$$175(X) = 100(5) + 75(15)$$

$$X = 9.286 \text{ cm.}$$

Example 3:

ABCD is a uniform rectangular lamina in which $AB = 36 \text{ cm}$ and $BC = 30 \text{ cm}$. E is the midpoint of BC. The triangle ABE is removed. Find the distance of the CoG of the remainder from AB and AD.

Sol:



The CoG of triangle ABE divides the median through A in the ratio 1:2. Its perpendicular distance from BE is $\frac{1}{3} \times 36 = 12$ cm. Similarly its perpendicular distance from AB = $\frac{1}{3} \times 15 = 5$ cm.

Figure	Area	Distance CoG from AB	Distance CoG from AD
ABCD	1080	15	18
ABE	270	5	24
AECD	810	x	y

Using: moment of whole = moment of removed part + moment of remainder

Taking moments about AB

$$1080(15) = 270(5) + 810(x)$$

$$x = 18\frac{1}{3} \text{ cm}$$

Taking moments about AD

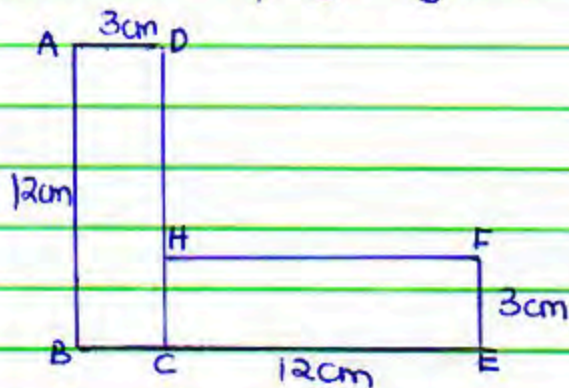
$$1080(18) = 270(24) + 810(y)$$

$$y = 16$$

N/B The C.o.g. of a uniform triangle is a $\frac{1}{3}$ of the distance ~~away~~^{up} from any base.

Exercises:

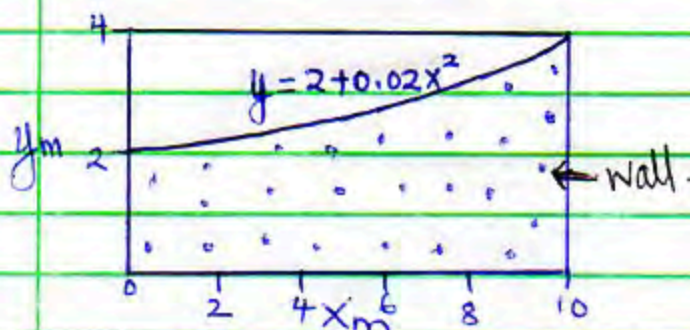
1. ABC is a uniform triangular lamina in which D and E are the mid-points of AB and AC. Find the centre of gravity of the quadrilateral BCED.
2. The figure below shows an L-shaped lamina formed by joining two uniform similar rectangles ABCD and CEFH. Given that the density of the first rectangle is twice that of the second, find the position of the centre of gravity.



$$\begin{pmatrix} x = 4\text{cm} \\ y = 4.5\text{cm} \end{pmatrix}$$

3. If the L-shaped lamina above is suspended from A, what will be the inclination of AB to the vertical? (28.07°).

4. An architect wants to build a wall with the profile shown in the figure. To estimate the effects of wind loads, he must determine the wall's area and the coordinates of its centroid. What are they?



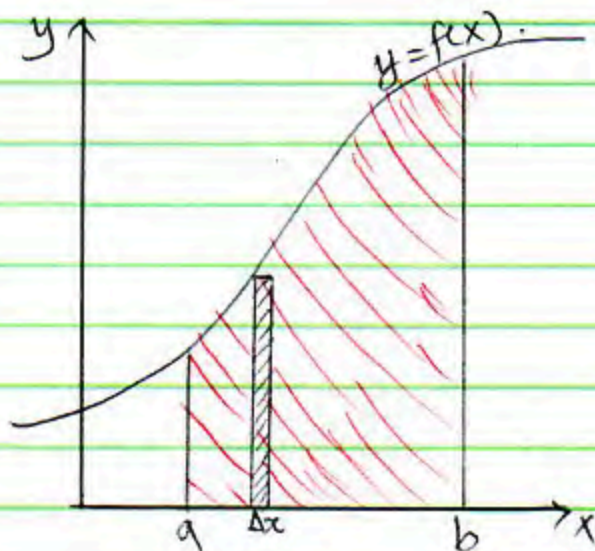
Ans: Area = 26.67 m²

$$\bar{x} = 5.63\text{ m}$$

$$\bar{y} = 1.40\text{ m}$$

Centroids for curved areas (2-D)

Consider the area defined by a function $y=f(x)$ and the vertical lines $x=a$ and $x=b$ as shown in the figure below.



To find the centroid, we sum the moments about the x and y axes of thin infinitesimally small strips (rectangles) as shown.

The typical rectangle above is x units from the y -axis and has width Δx and height $y=f(x)$.

Thus moment of area for each strip is $xf(x)\Delta x$.

Then the infinite sum of all such rectangles from a to b is $\int_a^b xf(x) dx$.

If A is the area of the shape (area under the curve from a to b), then

$$A\bar{x} = \int_a^b xf(x) dx$$

$$\text{i.e. } \bar{x} = \frac{1}{A} \int_a^b x f(x) dx.$$

Now, taking moments about the x-axis, the centroid of each thin strip is a distance $y/2 = \frac{f(x)}{2}$

from the x-axis. $\therefore$ moment of area of each strip is

$$\frac{f(x)}{2} f(x) dx.$$

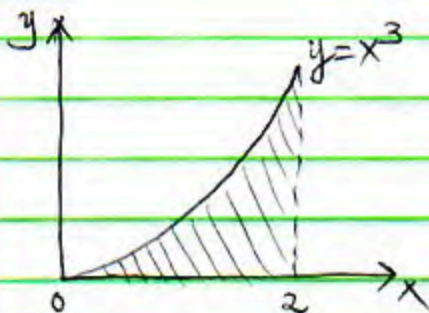
Hence the infinite sum of will be

$$\Rightarrow \bar{y} = \frac{1}{A} \int_a^b \frac{(f(x))^2}{2} dx.$$

Example 4:

Find the centroid of the area bounded by $y=x^3$, $x=2$ and the x-axis.

Sol:



$$\therefore y=f(x)=x^3, a=0 \text{ and } b=2.$$

We find the shaded area first:

$$A = \int_0^2 x^3 dx = \left[\frac{x^4}{4} \right]_0^2 = \frac{16}{4} = \underline{\underline{4}}$$

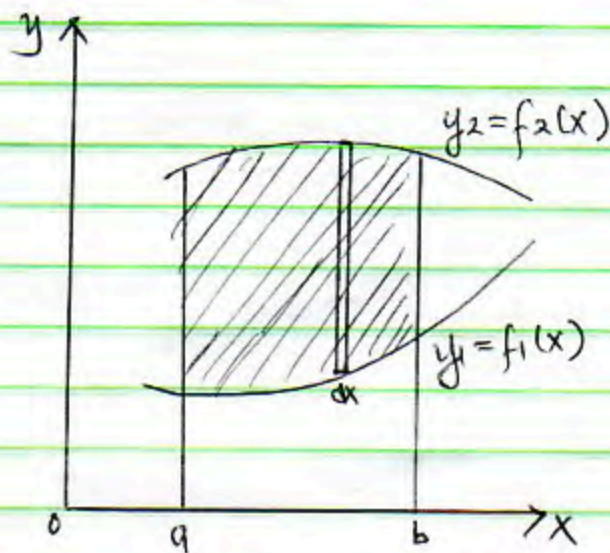
$$\text{then } \bar{x} = \frac{1}{A} \int_a^b x f(x) dx$$

$$\begin{aligned} &= \frac{1}{4} \int_0^2 x(x^3) dx = \frac{1}{4} \int_0^2 x^4 dx \\ &= \frac{1}{4} \left[\frac{x^5}{5} \right]_0^2 \\ &= \frac{32}{20} = 1.6 \end{aligned}$$

then the y coordinate,

$$\begin{aligned} \bar{y} &= \frac{1}{A} \int_a^b \frac{[f(x)]^2}{2} dx \\ &= \frac{1}{4} \int_0^2 \frac{(x^3)^2}{2} dx = \frac{1}{4} \int_0^2 \frac{x^6}{2} dx \\ &= \frac{1}{4} \left[\frac{x^7}{14} \right]_0^2 \\ &= \underline{\underline{2.29}} \end{aligned}$$

If the area is bounded by two curves as seen below,



The typical strip has width Δx and height $y_2 - y_1$ i.e. $f_2(x) - f_1(x)$, so the moment of area of each strip is $x (f_2(x) - f_1(x)) \Delta x$.

$$\therefore A\bar{x} = \int_a^b x [f_2(x) - f_1(x)] dx \Rightarrow \bar{x} = \frac{1}{A} \int_a^b x [f_2(x) - f_1(x)] dx$$

for y-coordinate

$$A\bar{y} = \int_a^b \frac{[y_2]^2 - [y_1]^2}{2} dx$$

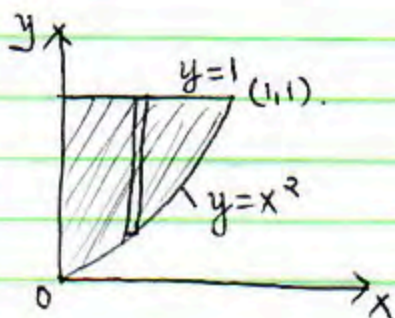
$$\text{i.e. } \bar{y} = \frac{1}{A} \int_a^b \frac{[f_2(x)]^2 - [f_1(x)]^2}{2} dx$$

$$\text{where } A = \int_a^b [f_2(x) - f_1(x)] dx$$

Example 5:

Find the centroid of the region bounded by the curve $y=x^2$, the y -axis and the line $y=1$.

Sol:



points of intersection

$$x^2=1$$

$x=-1$ or 1 . but region

is bounded by y -axis

$$\therefore x=1.$$

$$A = \int_a^b f_2(x) - f_1(x) dx$$

here $f_2(x) = y=1$.

$f_1(x) = y = x^2$.

$$= \int_0^1 (1-x^2) dx$$

$$= \left[x - \frac{x^3}{3} \right]_0^1 = \underline{\underline{\frac{2}{3}}}$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f_2(x) - f_1(x)] dx$$

$$= \frac{\frac{2}{3}}{\frac{2}{3}} \int_0^1 x (1-x^2) dx = \frac{3}{2} \int_0^1 x - x^3 dx$$

$$= \frac{3}{2} \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} \right) = \underline{\underline{\frac{3}{8}}}$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{[f_2(x)]^2 - [f_1(x)]^2}{2} dx$$

$$= \frac{3}{4} \int_0^1 (1-x^4) dx = \frac{3}{4} \left[x - \frac{x^5}{5} \right]_0^1$$

$$= \frac{3}{4} \left[1 - \frac{1}{5} \right] = \underline{\underline{\frac{3}{5}}}$$